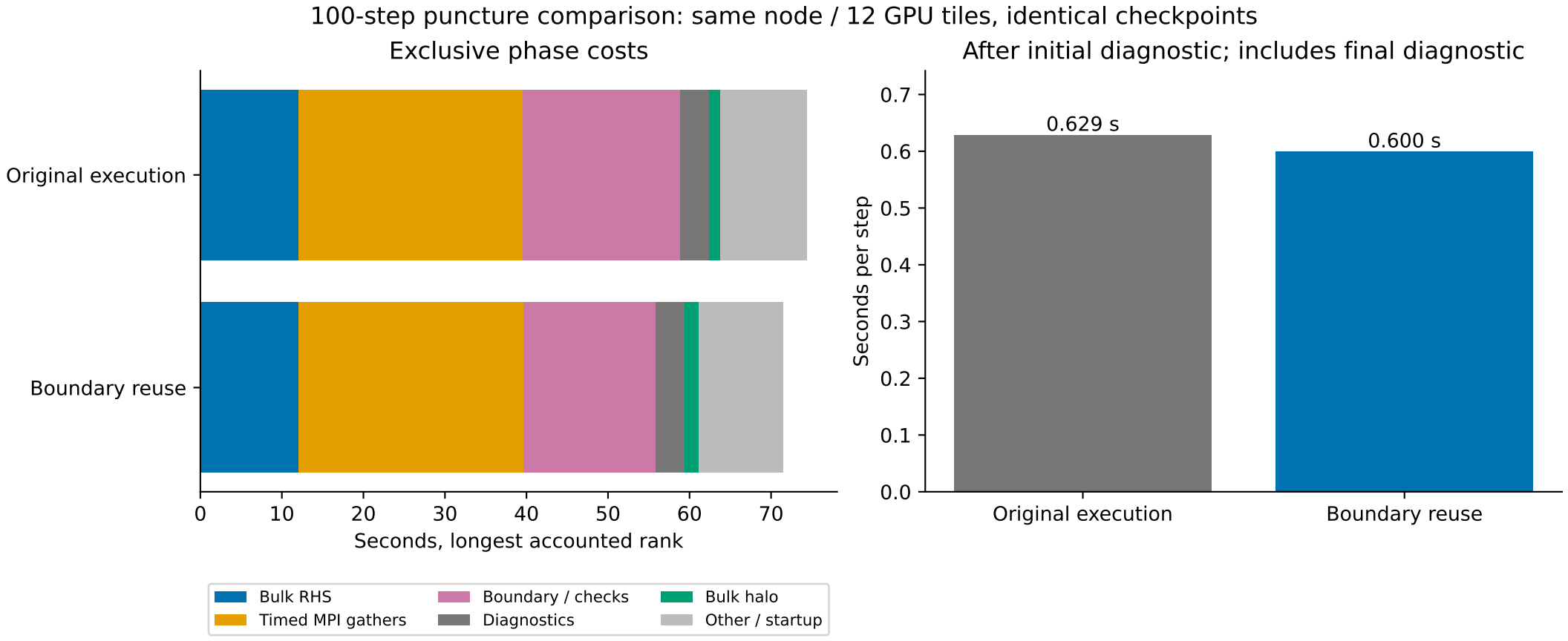
100-step puncture comparison: same node / 12 GPU tiles, identical checkpoints
Exclusive phase costs
Original execution
Boundary reuse
0
10
20
30
40
50
60
70
Seconds, longest accounted rank
Bulk RHS
Timed MPI gathers
Boundary / checks
Diagnostics
Bulk halo
Other / startup
After initial diagnostic; includes final diagnostic
Seconds per step
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.629 s
0.600 s
Original execution
Boundary reuse